

Sequences & Series



Arithmetic sequences: finding terms

- 1 Find the 12th term of the arithmetic sequence with $a = 5$, $d = 3$.
 - 2 Find the 20th term of the arithmetic sequence with $a = -4$, $d = 2$.
 - 3 Find the 15th term of the arithmetic sequence with $a = 100$, $d = -6$.
 - 4 Write the first four terms of the arithmetic sequence with $a = 7$, $d = -2$.
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Arithmetic sequences: finding a and d

- 5 The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find the first term a and common difference d .
 - 6 An arithmetic sequence has $a_1 = 10$ and $a_6 = 30$. Find the common difference d .
 - 7 An arithmetic sequence has $a_3 = 11$ and $a_7 = 27$. Find the common difference d .
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Arithmetic series: the sum formula

- 8 Find the sum of the first 20 terms of the arithmetic sequence with $a = 3$, $d = 4$, using $S_n = \frac{n}{2}(2a + (n - 1)d)$.
 - 9 Find the sum of the first 15 terms of the arithmetic sequence with $a = 50$, $d = -3$.
 - 10 Find the sum of the arithmetic series $2 + 5 + 8 + \cdots + 50$.
 - 11 Find the sum of the first 10 positive multiples of 6.
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Geometric sequences: finding terms

- 12 Find the 6th term of the geometric sequence with $a = 3$, $r = 2$.
- 13 Find the 5th term of the geometric sequence with $a = 100$, $r = 0.5$.
- 14 Find the 4th term of the geometric sequence with $a = 2$, $r = -3$.

- 15 Write the first four terms of the geometric sequence with $a = 5$, $r = 3$.
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Geometric sequences: finding a and r

- 16 The 2nd term of a geometric sequence is 6 and the 5th term is 162. Find r and a .
- 17 A geometric sequence has $a_1 = 4$ and $a_4 = 108$. Find r .
- 18 A geometric sequence has 3rd term 12 and 6th term 96. Find a and r .
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Geometric series: the finite sum

- 19 Find the sum of the first 6 terms of the geometric series with $a = 3$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.
- 20 Find the sum of the first 5 terms of the geometric series with $a = 100$, $r = 0.5$.
- 21 Find the sum of the first 4 terms of the geometric series with $a = 2$, $r = -3$.
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Infinite geometric series

- 22 Find the sum of the infinite geometric series with $a = 4$, $r = \frac{1}{2}$.
- 23 Find the sum of the infinite geometric series with $a = 9$, $r = \frac{1}{3}$.
- 24 Find the sum of the infinite geometric series $8 - 4 + 2 - 1 + \dots$.
- 25 Express the repeating decimal $0.333\dots$ as a fraction using an infinite geometric series with $a = \frac{3}{10}$, $r = \frac{1}{10}$.
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Sigma notation

- 26 Evaluate $\sum_{k=1}^5 (2k + 1)$.
- 27 Evaluate $\sum_{k=1}^4 3^k$.
- 28 Use $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{10} k$.
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Recursive sequences

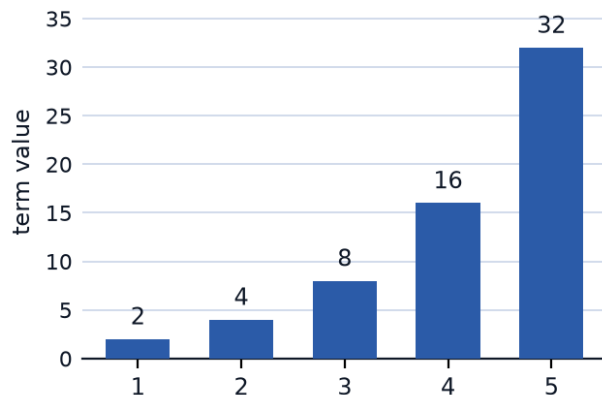
- 29** A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n + 5$. Find a_5 .
- 30** A sequence satisfies $a_1 = 3$ and $a_{n+1} = 2a_n - 1$. Find a_4 .
- 31** A sequence satisfies $a_1 = 1$, $a_2 = 1$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. Find a_6 .
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Identifying arithmetic and geometric sequences

- 32** State whether the sequence 4, 9, 14, 19, ... is arithmetic or geometric, and give the common difference or ratio.
- 33** State whether the sequence 2, 6, 18, 54, ... is arithmetic or geometric, and give the common difference or ratio.
- 34** State whether the sequence 100, 90, 80, 70, ... is arithmetic or geometric, and find its 10th term.
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Reading a sequence from its terms

- 35** The bar chart shows the first five terms of a sequence.



- a** Determine whether the sequence is arithmetic or geometric, and find the common ratio or difference.
- b** Find the 8th term.
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Word problems: sequences in context

- 36 This question has two parts. An employee's starting monthly salary is 4000 dollars, and it increases by a fixed amount of 150 dollars each month.
- a Write a formula for the salary a_n in month n (with $a_1 = 4000$).
 - b Find the salary in month 12.
- 37 This question has two parts. A bacteria culture starts with 500 cells and triples every hour. Let a_n be the number of cells after n hours, with $a_0 = 500$.
- a Write a formula for a_n .
 - b Find the number of cells after 4 hours.
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MCQ — Arithmetic sequences: finding terms

- 38 Find the 14th term of the arithmetic sequence with $a = 6$, $d = 4$.
- a. 62
 - b. 54
 - c. 50
 - d. 58
- 39 Find the 18th term of the arithmetic sequence with $a = -7$, $d = 3$.
- a. 50
 - b. 41
 - c. 47
 - d. 44
- 40 Find the 12th term of the arithmetic sequence with $a = 80$, $d = -5$.
- a. 20
 - b. 30
 - c. 25
 - d. -25

41 Write the first four terms of the arithmetic sequence with $a = 9$, $d = -3$.

a. 9, 12, 15, 18

b. 9, 6, 3, 0

c. 9, 3, 0, -3

d. 9, -3 , -15 , -27

42 Find the 25th term of the arithmetic sequence with $a = 2$, $d = 7$.

a. 177

b. 170

c. 175

d. 163

MCQ — Arithmetic sequences: finding a and d

43 The 5th term of an arithmetic sequence is 19 and the 10th term is 44. Find the first term a and common difference d .

a. $a = -5$, $d = 5$

b. $a = -1$, $d = 4$

c. $a = 4$, $d = 5$

d. $a = -1$, $d = 5$

44 An arithmetic sequence has $a_1 = 8$ and $a_7 = 32$. Find the common difference d .

a. $d = 4$

b. $d = 24$

c. $d = 4.8$

d. $d = 3$

- 45 An arithmetic sequence has $a_2 = 9$ and $a_6 = 29$. Find the common difference d .
- a. $d = 5.5$
 - b. $d = 4$
 - c. $d = 20$
 - d. $d = 5$
- 46 The 3rd term of an arithmetic sequence is 15 and the 8th term is 45. Find the first term a and common difference d .
- a. $a = 3, d = 5$
 - b. $a = 3, d = 6$
 - c. $a = 6, d = 3$
 - d. $a = 9, d = 6$
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MCQ — Arithmetic series: the sum formula

- 47 Find the sum of the first 25 terms of the arithmetic sequence with $a = 4, d = 3$, using $S_n = \frac{n}{2}(2a + (n - 1)d)$.
- a. 1,075
 - b. 960
 - c. 1,000
 - d. 925

- 48 Find the sum of the first 18 terms of the arithmetic sequence with $a = 60$, $d = -4$.
- a. 540
 - b. 456
 - c. 468
 - d. 480
- 49 Find the sum of the arithmetic series $3 + 7 + 11 + \cdots + 59$.
- a. 450
 - b. 434
 - c. 465
 - d. 480
- 50 Find the sum of the first 12 positive multiples of 7.
- a. 504
 - b. 546
 - c. 462
 - d. 588
- 51 Find the sum of the first 30 terms of the arithmetic sequence with $a = 1$, $d = 2$ (the odd numbers).
- a. 960
 - b. 930
 - c. 870
 - d. 900

MCQ — Geometric sequences: finding terms

- 52 Find the 7th term of the geometric sequence with $a = 2$, $r = 3$.
- a. 1,458
 - b. 729
 - c. 4,374
 - d. 486
- 53 Find the 6th term of the geometric sequence with $a = 200$, $r = 0.5$.
- a. 25
 - b. 3.125
 - c. 12.5
 - d. 6.25
- 54 Find the 5th term of the geometric sequence with $a = 3$, $r = -2$.
- a. 24
 - b. 48
 - c. -24
 - d. -48
- 55 Write the first four terms of the geometric sequence with $a = 4$, $r = 2$.
- a. 4, 6, 8, 10
 - b. 4, 16, 32, 64
 - c. 4, 8, 12, 16
 - d. 4, 8, 16, 32

MCQ — Geometric sequences: finding a and r

56 The 2nd term of a geometric sequence is 8 and the 5th term is 64. Find r and a .

a. $r = 2, a = 8$

b. $r = 4, a = 2$

c. $r = 2, a = 4$

d. $r = 8, a = 1$

57 A geometric sequence has $a_1 = 5$ and $a_4 = 135$. Find r .

a. $r = 9$

b. $r = 27$

c. $r = 3$

d. $r = 2$

58 A geometric sequence has 3rd term 18 and 6th term 486. Find a and r .

a. $a = 18, r = 3$

b. $a = 6, r = 3$

c. $a = 2, r = 3$

d. $a = 2, r = 9$

59 A geometric sequence has 2nd term 4 and 5th term -108 . Find r .

a. $r = 3$

b. $r = -27$

c. $r = -3$

d. $r = -9$

60 A geometric sequence has $a_1 = 6$ and $a_4 = -162$. Find r .

- a. $r = 3$
- b. $r = -9$
- c. $r = -3$
- d. $r = -27$

MCQ — Geometric series: the finite sum

61 Find the sum of the first 7 terms of the geometric series with $a = 2$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

- a. 256
- b. 508
- c. 254
- d. 127

62 Find the sum of the first 6 terms of the geometric series with $a = 200$, $r = 0.5$.

- a. 196.875
- b. 400
- c. 396.875
- d. 393.75

63 Find the sum of the first 5 terms of the geometric series with $a = 3$, $r = -2$.

- a. 93
- b. 63
- c. -33
- d. 33

64 Find the sum of the first 8 terms of the geometric series with $a = 1$, $r = 2$.

- a. 255
- b. 128
- c. 254
- d. 256

MCQ — Infinite geometric series

65 Find the sum of the infinite geometric series with $a = 6$, $r = \frac{1}{3}$.

- a. 18
- b. 6.75
- c. 8
- d. 9

66 Find the sum of the infinite geometric series with $a = 12$, $r = \frac{1}{4}$.

- a. 14
- b. 16
- c. 48
- d. 15

67 Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \dots$.

- a. 4.5
- b. 9
- c. 6
- d. 6.75

- 68 Express the repeating decimal $0.777\ldots$ as a fraction using an infinite geometric series with $a = 0.7$, $r = 0.1$.

- a. $\frac{7}{90}$
- b. $\frac{77}{100}$
- c. $\frac{7}{9}$
- d. $\frac{7}{10}$

- 69 Find the sum of the infinite geometric series with $a = 10$, $r = \frac{2}{5}$.

- a. $\frac{50}{3}$
- b. $\frac{25}{3}$
- c. 14
- d. 25

MCQ — Sigma notation

- 70 Evaluate $\sum_{k=1}^6 (3k - 1)$.

- a. 51
- b. 54
- c. 57
- d. 63

- 71 Evaluate $\sum_{k=1}^5 2^k$.

- a. 60
- b. 62
- c. 63
- d. 32

72 Use $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{15} k$.

a. 120

b. 110

c. 105

d. 136

73 Evaluate $\sum_{k=2}^5 k^2$.

a. 55

b. 30

c. 50

d. 54

MCQ — Recursive sequences

74 A sequence satisfies $a_1 = 3$ and $a_{n+1} = a_n + 7$. Find a_5 .

a. 31

b. 38

c. 28

d. 24

75 A sequence satisfies $a_1 = 4$ and $a_{n+1} = 3a_n - 2$. Find a_4 .

a. 10

b. 244

c. 28

d. 82

- 76 A sequence satisfies $a_1 = 1$, $a_2 = 2$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. Find a_7 .
- a. 21
 - b. 34
 - c. 8
 - d. 13
- 77 A sequence satisfies $a_1 = 2$ and $a_{n+1} = 2a_n + 1$. Find a_4 .
- a. 11
 - b. 23
 - c. 5
 - d. 47
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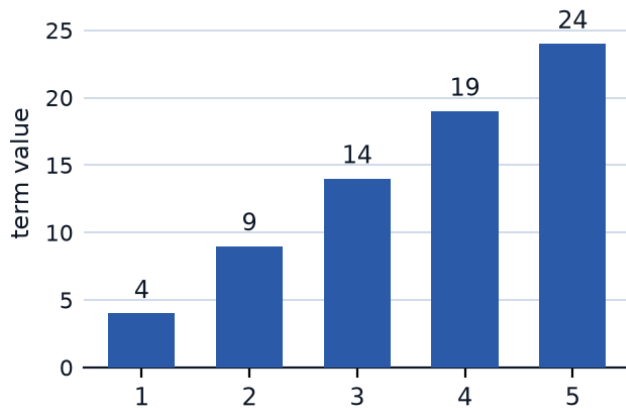
MCQ — Identifying arithmetic and geometric sequences

- 78 State whether 6, 11, 16, 21, ... is arithmetic or geometric, and give the common difference or ratio.
- a. Arithmetic, $d = 5$
 - b. Geometric, $r = \frac{11}{6}$
 - c. Geometric, $r = 5$
 - d. Arithmetic, $d = 6$
- 79 State whether 3, 9, 27, 81, ... is arithmetic or geometric, and give the common difference or ratio.
- a. Geometric, $r = 6$
 - b. Arithmetic, $d = 6$
 - c. Arithmetic, $d = 3$
 - d. Geometric, $r = 3$

- 80 State whether 90, 80, 70, 60, ... is arithmetic or geometric, and find its 12th term.
- a. Arithmetic, -20
 - b. Arithmetic, -10
 - c. Arithmetic, 30
 - d. Geometric, -20
- 81 State whether 5, 10, 20, 40, ... is arithmetic or geometric, and give the common difference or ratio.
- a. Geometric, $r = 2$
 - b. Geometric, $r = 5$
 - c. Arithmetic, $d = 2$
 - d. Arithmetic, $d = 5$

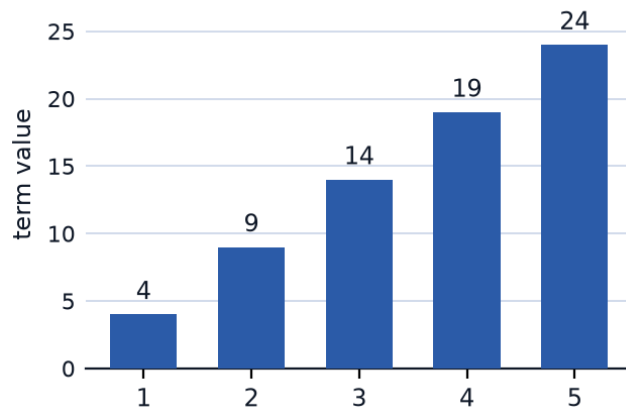
MCQ — Reading a sequence from its terms

- 82 The bar chart shows the first five terms of a sequence. Find the common difference.



- a. 6
- b. 5
- c. 4
- d. 9

83 Using the same bar chart, predict the 8th term.



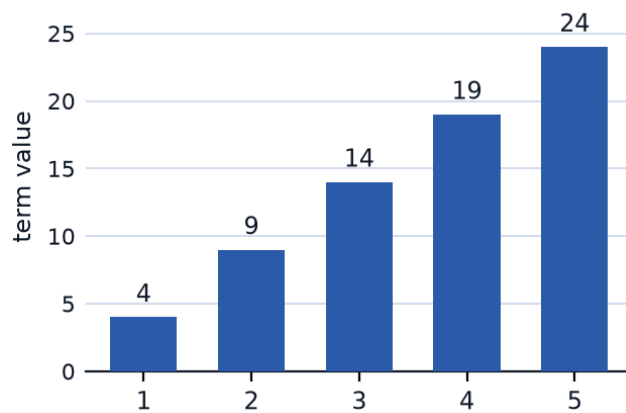
a. 29

b. 34

c. 39

d. 44

84 Using the same bar chart, find the sum of the first five terms shown.



a. 70

b. 66

c. 24

d. 75

MCQ — Word problems: sequences in context

- 85** An employee's starting monthly salary is 3,500 dollars, and it increases by 150 dollars each year. Find the salary in year 6.
- a. 3,950 dollars
 - b. 4,400 dollars
 - c. 4,100 dollars
 - d. 4,250 dollars
- 86** A bacteria culture starts with 300 cells and doubles every hour. Find the number of cells after 5 hours.
- a. 19,200
 - b. 1,500
 - c. 9,600
 - d. 4,800
- 87** A theater's ticket sales grow according to a geometric sequence: 200 tickets sold on day 1, with a common ratio of 1.5. Find the number of tickets sold on day 4.
- a. 900
 - b. 675
 - c. 300
 - d. 450

Multiple choice — easy

88 Find the 9th term of the arithmetic sequence with $a = -10$, $d = 2$.

a. 4

b. 9

c. 8

d. 6

89 Find the 16th term of the arithmetic sequence with $a = 20$, $d = -7$.

a. -82

b. -92

c. -85

d. -78

90 Find the 14th term of the arithmetic sequence with $a = 6$, $d = 2$.

a. 35

b. 32

c. 34

d. 30

91 Find the 5th term of the arithmetic sequence with $a = -19$, $d = -5$.

a. -36

b. -39

c. -44

d. -34

92 Find the 16th term of the arithmetic sequence with $a = -14$, $d = 5$.

a. 56

b. 61

c. 66

d. 64

93 Find the 13th term of the arithmetic sequence with $a = -16$, $d = 9$.

a. 95

b. 101

c. 83

d. 92

94 Find the 20th term of the arithmetic sequence with $a = -7$, $d = -7$.

a. -133

b. -140

c. -147

d. -137

95 Find the 10th term of the arithmetic sequence with $a = 12$, $d = 5$.

a. 60

b. 62

c. 52

d. 57

- 96 Find the 7th term of the arithmetic sequence with $a = 11$, $d = -5$.
- a. -14
 - b. -19
 - c. -24
 - d. -16
- 97 Find the 8th term of the arithmetic sequence with $a = 6$, $d = 2$.
- a. 23
 - b. 22
 - c. 18
 - d. 20
- 98 Find the 15th term of the arithmetic sequence with $a = -20$, $d = -8$.
- a. -129
 - b. -132
 - c. -124
 - d. -140
- 99 Find the 10th term of the arithmetic sequence with $a = 12$, $d = -6$.
- a. -42
 - b. -48
 - c. -39
 - d. -36

100 Find the 11th term of the arithmetic sequence with $a = -8$, $d = 1$.

- a. 1
- b. 3
- c. 2
- d. 5

101 Find the 17th term of the arithmetic sequence with $a = -4$, $d = 5$.

- a. 71
- b. 81
- c. 76
- d. 79

102 Find the 5th term of the arithmetic sequence with $a = -2$, $d = 9$.

- a. 25
- b. 43
- c. 37
- d. 34

Multiple choice — medium

103 Find the 7th term of the geometric sequence with $a = -2$, $r = 2$.

- a. -131
- b. -128
- c. 128
- d. -125

104 Find the 3rd term of the geometric sequence with $a = -2$, $r = 2$.

- a. -8
- b. 8
- c. -11
- d. -5

105 Find the 4th term of the geometric sequence with $a = 3$, $r = -3$.

- a. -81
- b. 81
- c. -78
- d. -84

106 Find the 6th term of the geometric sequence with $a = -9$, $r = -2$.

- a. 291
- b. -288
- c. 285
- d. 288

107 Find the 4th term of the geometric sequence with $a = -7$, $r = -2$.

- a. 56
- b. 59
- c. -56
- d. 53

108 Find the 6th term of the geometric sequence with $a = -6$, $r = 2$.

- a. -195
- b. 192
- c. -192
- d. -189

109 Find the 5th term of the geometric sequence with $a = 1$, $r = 3$.

- a. 78
- b. 81
- c. -81
- d. 84

110 Find the 4th term of the geometric sequence with $a = 1$, $r = -3$.

- a. -27
- b. -30
- c. 27
- d. -24

111 Find the 3rd term of the geometric sequence with $a = -4$, $r = 2$.

- a. -19
- b. -13
- c. 16
- d. -16

112 Find the 5th term of the geometric sequence with $a = -2$, $r = 3$.

- a. -162
- b. -159
- c. 162
- d. -165

113 Find the 7th term of the geometric sequence with $a = -2$, $r = -3$.

- a. 1458
- b. -1461
- c. -1455
- d. -1458

114 Find the 3rd term of the geometric sequence with $a = -5$, $r = -3$.

- a. -48
- b. -42
- c. -45
- d. 45

115 Find the 7th term of the geometric sequence with $a = -7$, $r = -2$.

- a. 448
- b. -445
- c. -448
- d. -451

116 Find the 6th term of the geometric sequence with $a = -8$, $r = 2$.

- a. 256
- b. -253
- c. -259
- d. -256

117 Find the 3rd term of the geometric sequence with $a = -2$, $r = 3$.

- a. -15
- b. 18
- c. -21
- d. -18

Multiple choice — hard

118 Find the sum of the infinite geometric series with $a = 4$, $r = \frac{1}{2}$.

- a. 8
- b. 6
- c. -8
- d. 10

119 Find the sum of the infinite geometric series with $a = 18$, $r = \frac{1}{3}$.

- a. 25
- b. 29
- c. 27
- d. -27

120 Find the sum of the infinite geometric series with $a = 6$, $r = \frac{1}{3}$.

- a. 9
- b. 7
- c. 11
- d. -9

121 Find the sum of the infinite geometric series with $a = 16$, $r = \frac{1}{4}$.

- a. $\frac{64}{3}$
- b. $-\frac{64}{3}$
- c. $\frac{58}{3}$
- d. $\frac{70}{3}$

122 Find the sum of the infinite geometric series with $a = 30$, $r = \frac{1}{5}$.

- a. $\frac{79}{2}$
- b. $\frac{75}{2}$
- c. $-\frac{75}{2}$
- d. $\frac{71}{2}$

123 Evaluate $\sum_{k=1}^5 (k + 4)$.

- a. 31
- b. -35
- c. 39
- d. 35

124 Evaluate $\sum_{k=1}^6 (k - 2)$.

a. -9

b. 13

c. 9

d. 5

125 Evaluate $\sum_{k=1}^6 (2k + 2)$.

a. -54

b. 54

c. 50

d. 58

126 Evaluate $\sum_{k=1}^5 (2k + 2)$.

a. -40

b. 36

c. 40

d. 44

127 Evaluate $\sum_{k=1}^6 (k + 5)$.

a. 55

b. 51

c. -51

d. 47

128 A sequence satisfies $a_1 = 6$ and $a_{n+1} = a_n + 2$. Find a_6 .

- a. 14
- b. 13
- c. 16
- d. 19

129 A sequence satisfies $a_1 = 6$ and $a_{n+1} = 3a_n + 3$. Find a_4 .

- a. 201
- b. 198
- c. 66
- d. 204

130 A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n + 4$. Find a_4 .

- a. 10
- b. 14
- c. 17
- d. 11

131 A sequence satisfies $a_1 = 4$ and $a_{n+1} = 3a_n - 1$. Find a_6 .

- a. 854
- b. 284
- c. 851
- d. 848

132 A sequence satisfies $a_1 = 7$ and $a_{n+1} = a_n + 3$. Find a_6 .

a. 22

b. 27

c. 25

d. 19